String Theory and M-Theory

Lecture 5: Scales, Strings, and Twenty-Six Dimensions

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Companion notes curated by LazyingArt LLC with Video2Book

Chapter 1

Scales, Strings, and Twenty-Six Dimensions

Overview. What sets the spacing of a string spectrum? The answer is the tension, a quantity with dimensions of mass squared. For fundamental strings that observation leads toward Planck units, the practical impossibility of direct excitation, and the onset of strong gravity. After a deliberate break, we return to the bosonic open-string ground state. Regulating its zero-point energy gives the historical *easy* route to twenty-four transverse directions and hence twenty-six spacetime dimensions.

1.1 The Scale We Had Left Out

So far we have spoken about vibrating and rotating strings without saying what one unit of excitation means. A hadron can be pictured as a string, and its oscillators can be excited, but how large is the jump from one state to the next? What determines it?

There is only one dimensional parameter in the elementary string model. To find it, begin with an ordinary stretched spring. If its endpoints are separated by an amount ℓ, Hooke's law gives

$$E_{\text{spring}} = \frac{k\ell^2}{2}. \tag{1.1}$$

The factor of one-half was supplied by the class after the first version was written. Nothing essential will depend on such numerical factors in this opening estimate.

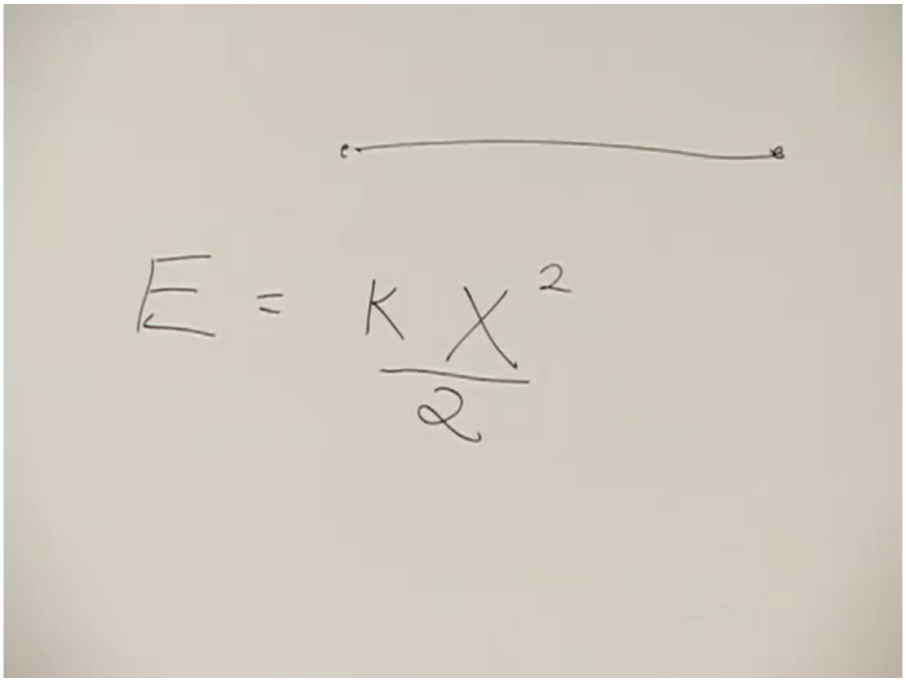


Figure 1.1: The stretched segment and its Hooke-law potential energy.
Lecture frame: 00:02:28.

The energy in (1.1) is nonrelativistic. In the infinite-momentum description used in the preceding lectures, internal nonrelativistic energy is proportional not to m, but to m^2. We therefore write schematically

$$m^2 \sim k\ell^2, \qquad m \sim \sqrt{k}\,\ell. \tag{1.2}$$

In the string's rest frame its mass is its energy, so the second relation has the form

$$E_{\text{rest}} \sim T_s \ell, \qquad T_s \equiv \sqrt{k}. \tag{1.3}$$

The coefficient T_s is the string tension: the energy stored per unit length.

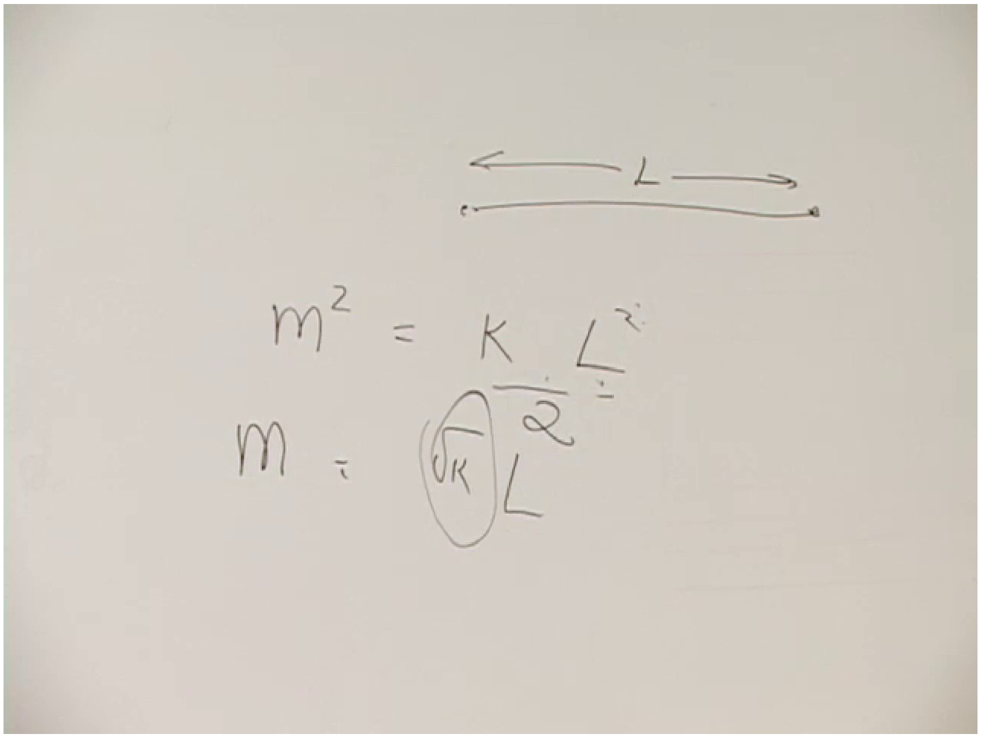


Figure 1.2: The board passes from $m^2 \sim k\ell^2$ to a mass proportional to length.
Lecture frame: 00:05:42.

This notation is adapted to the lecture's spring analogy. In standard relativistic string notation one usually writes

$$T_s = \frac{1}{2\pi\alpha'}, \tag{1.4}$$

so increments of string level produce increments of m^2 of order $1/\alpha'$, equivalently of order the tension. [1]

1.1.1 Dimensions and oscillator gaps

Set $c = \hbar = 1$. Energy then has dimensions of inverse length:

$$[E] = L^{-1}. \tag{1.5}$$

Since tension is energy per unit length,

$$[T_s] = L^{-2} = E^2. \tag{1.6}$$

That is why a string spectrum is naturally spaced in mass squared.

[1] Editorial clarification: The lecture's effective k is chosen so that $\sqrt{k}$ plays the role of tension. The convention-independent statement is that T_s has dimensions of mass squared and that the Regge slope satisfies $\alpha' \propto T_s^{-1}$.

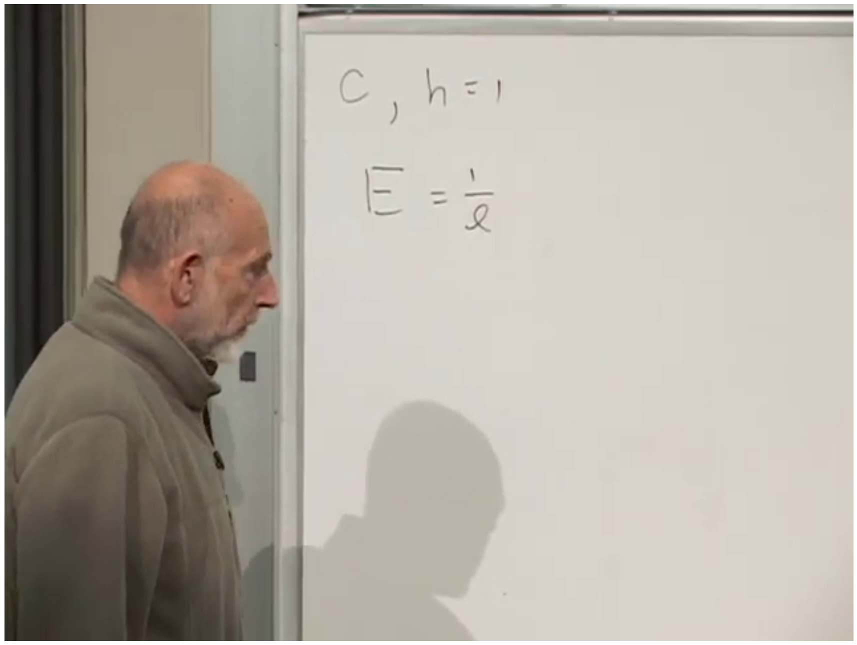


Figure 1.3: Natural units on the board: $c = \hbar = 1$, energy is inverse length, and the oscillator frequency is controlled by the spring scale.

Lecture frame: 00:06:05.

The ordinary oscillator makes the same point from another direction. For a spring with mass μ,

$$\omega = \sqrt{\frac{k}{\mu}}, \qquad \Delta E = \hbar\omega. \tag{1.7}$$

A stiffer spring has a higher frequency and a larger quantum gap. A string contains infinitely many such normal modes, but their common scale is fixed by T_s. Once the tension is known, the spacing of the string spectrum is known.

Tension also has the dimensions of force, because force times distance is energy. Imagine anchoring one endpoint and suspending a weight from the other. An ideal relativistic string with $E = T_s\ell$ can support a force set by T_s, independent of its length. The hadronic tension inferred from observed excitation energies corresponds, in this deliberately fanciful comparison, to the weight of a truck. No truck has ever been hung from a meson; the measured spectrum gives the tension, and the tension gives the mechanical comparison.

Fundamental strings would be far stiffer. If they underlie electrons, photons, and gravitons, their first oscillator excitations must lie far above observed particle energies. In the same fanciful gravitational experiment their tension could support something of galactic mass. The point of the comparison is not engineering. It is that an enormous tension means both a large excitation energy and a very small ground-state fluctuation.

1.2 Units Made from Universal Constants

What scale is enormous enough? The natural benchmark is the Planck scale. Rather than quote it, let us construct it.

Meters, kilograms, and seconds are excellent laboratory units, but their origins are human-sized. A meter is convenient for cloth or rope; a foot is named after a body part; a second is a manageable fraction of a day. One could set the proton radius equal to one and simplify some nuclear formulas, but the proton is only one particular composite particle. Such a choice would not simplify all of physics.

Three dimensional constants have a more universal reach:

- c sets the limiting speed for every material system;
- $\hbar$ appears in the uncertainty relations of every quantum system;
- G couples gravity to every form of mass-energy.

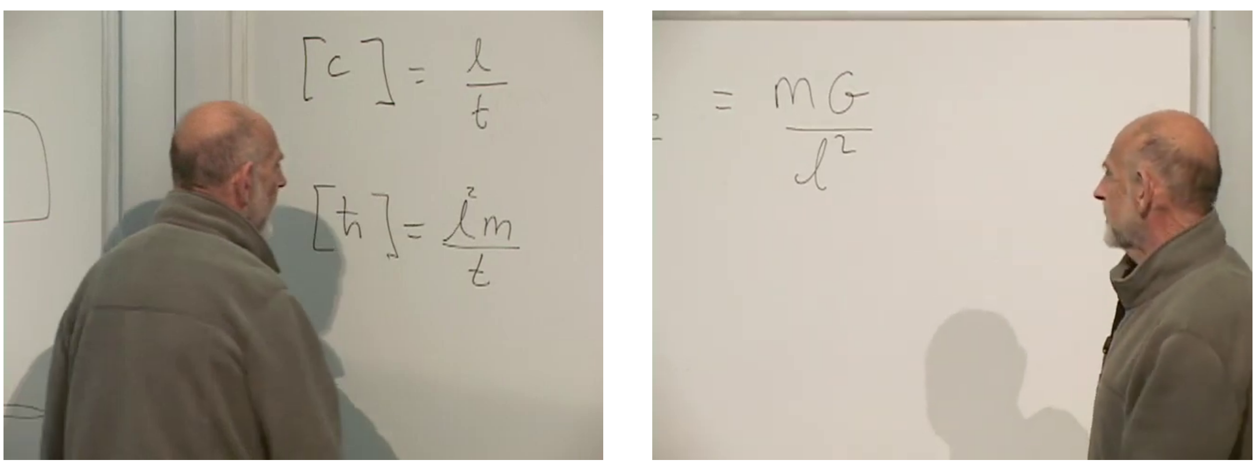
Choosing units in which $c = \hbar = G = 1$ is equivalent to constructing units of length, time, and mass from these constants.

Their ordinary dimensions are

$$\begin{aligned} [c] &= LT^{-1}, \qquad [\hbar] = ML^2T^{-1}, \\ [G] &= L^3M^{-1}T^{-2}. \end{aligned} \tag{1.8}$$

The middle relation follows directly from $\Delta x\,\Delta p \gtrsim \hbar$. The last follows, for example, from

$$a = \frac{GM}{r^2}. \tag{1.9}$$



(a) Units of c and $\hbar$.

(b) Units of G from Newtonian acceleration.

Figure 1.4: The dimensional-analysis ingredients assembled in class.

Lecture frame: 00:25:43–00:26:40.

1.2.1 The Planck area first

Seek powers p, q, r for which

$$G^p \hbar^q c^r \tag{1.10}$$

has dimensions L^2. Choosing area first avoids an early square root and also anticipates the special role that area will later play in gravity. From (1.8), cancellation of mass and time gives

$$-p + q = 0, \qquad -2p - q - r = 0. \tag{1.11}$$

The remaining length power must satisfy

$$3p + 2q + r = 2. \tag{1.12}$$

Thus $q = p$, $r = -3p$, and $p = 1$. The Planck area is

$$\ell_P^2 = \frac{G\hbar}{c^3}, \tag{1.13}$$

and consequently

$$\ell_P = \sqrt{\frac{G\hbar}{c^3}}, \qquad t_P = \frac{\ell_P}{c}, \qquad m_P = \sqrt{\frac{\hbar c}{G}}. \tag{1.14}$$

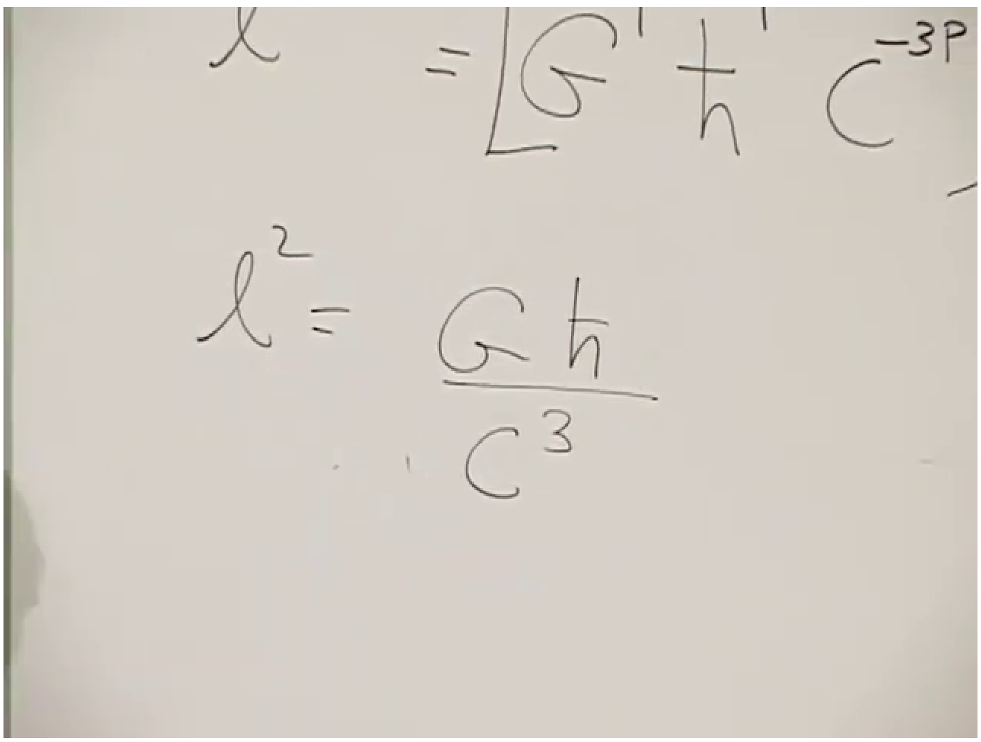


Figure 1.5: The completed dimensional argument for the Planck area and length.
Lecture frame: 00:31:34.

At order-of-magnitude precision,

$$\begin{aligned} \ell_P &\sim 10^{-35}\,\text{m}, \qquad t_P \sim 10^{-43}\,\text{s}, \\ m_P &\sim 10^{-8}\,\text{kg}, \quad m_P c^2 \sim 10^{19}\,\text{GeV}. \end{aligned} \tag{1.15}$$

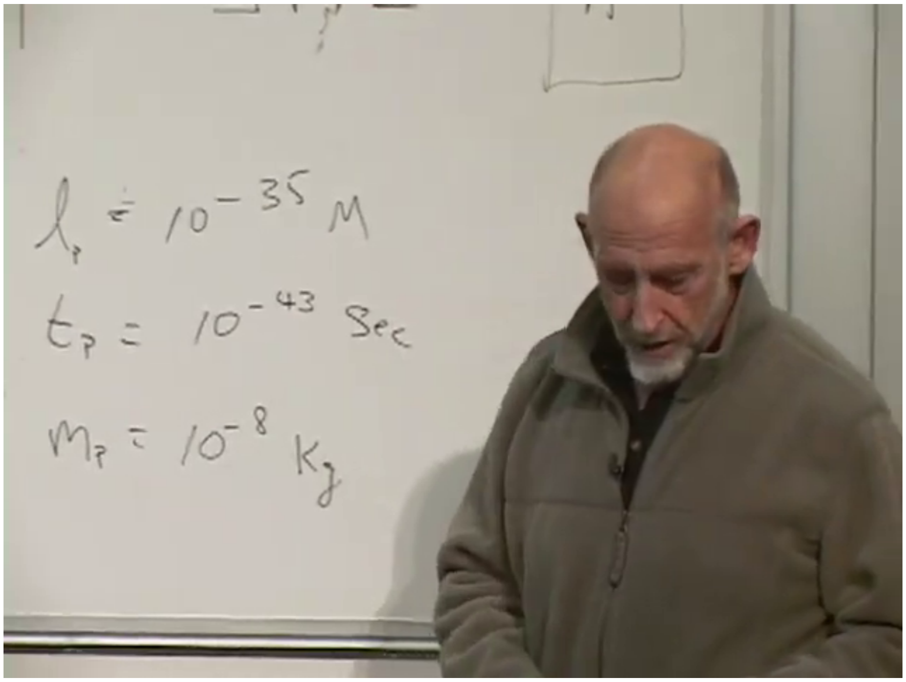


Figure 1.6: The three Planck scales recorded to order of magnitude.
Lecture frame: 00:35:12.

The length and time are unimaginably small, while the mass is surprisingly macroscopic by particle standards, roughly the mass of a fine dust grain. Its rest energy is of order a few gigajoules, comparable to the chemical energy in a tank of gasoline. The observable universe, by contrast, spans roughly 10^{60} Planck lengths and has lived for roughly 10^{60} Planck times. These comparisons are meant to give the units physical weight, not to improve their numerical precision.

If the characteristic string length $\ell_s = \sqrt{\alpha'}$ is near ℓ_P, then the string's quantum size is near ℓ_P, its oscillation period is near t_P, and its first massive excitation is near m_P. The lecture allows factors of ten, one hundred, or more; the identification is an expectation, not a dimensional identity.

Question from the audience. Is the Planck-scale mass the increase above the particle's ground state? [a]

Response. The evenly spaced quantity is mass squared. If the ground state is light, the first string excitation has m^2 larger by an amount of order m_P^2, so its mass and required center-of-mass energy are of order m_P.

[a] Lecture timestamp: 00:38:37.

1.3 How Close Should the Two Scales Be?

Question from the audience. Dimensional analysis gives ℓ_P, but why should that number be the length of a string? [a]

Response. It is a physical guess supported by gravity and by the absence of observed string structure, not a consequence of dimensional analysis alone. The string length may be larger than the Planck length, which means its energy scale is lower. The lecture argues only that it should not be an ordinary particle-physics length and that pushing it far below the Planck length runs into gravitational collapse.

[a] Lecture timestamp: 00:40:21.

The relation between the two scales is in fact coupling dependent. In a complete compactification, ℓ_P, ℓ_s, the string coupling, and the compactification volume enter together. There is no universal equation $\ell_s = \ell_P$. What survives from the lecture is the scale argument: string effects must enter very high, and sub-Planckian localization cannot be discussed while neglecting gravity. [2]

Question from the audience. If the Planck length is treated as a smallest useful length, is the Planck mass also the smallest possible mass? [a]

Response. No. Ordinary particles are far lighter. The Planck mass is the scale at which a compact object can have a gravitational radius comparable to its quantum wavelength. In the lecture's heuristic language, it is the scale of the lightest semiclassical black hole, with a lifetime of order a Planck time; the last statement should not be mistaken for a precise description of a fully quantum object.

[a] Lecture timestamp: 00:43:27.

1.3.1 What running couplings can tell us

There is indirect evidence that conventional quantum field theory remains valid to scales far beyond direct accelerator reach. The three standard-model gauge couplings are measured at accessible energies, and their scale dependence can be extrapolated. In the lecture's schematic plot, the $U(1)$, $SU(2)$, and $SU(3)$ couplings approach one another at a high unification scale, perhaps a few orders of magnitude below the Planck energy.

[2] Editorial clarification: Modern string constructions need not identify the string length exactly with the four-dimensional Planck length. Their relation depends on the string coupling and compactification data.

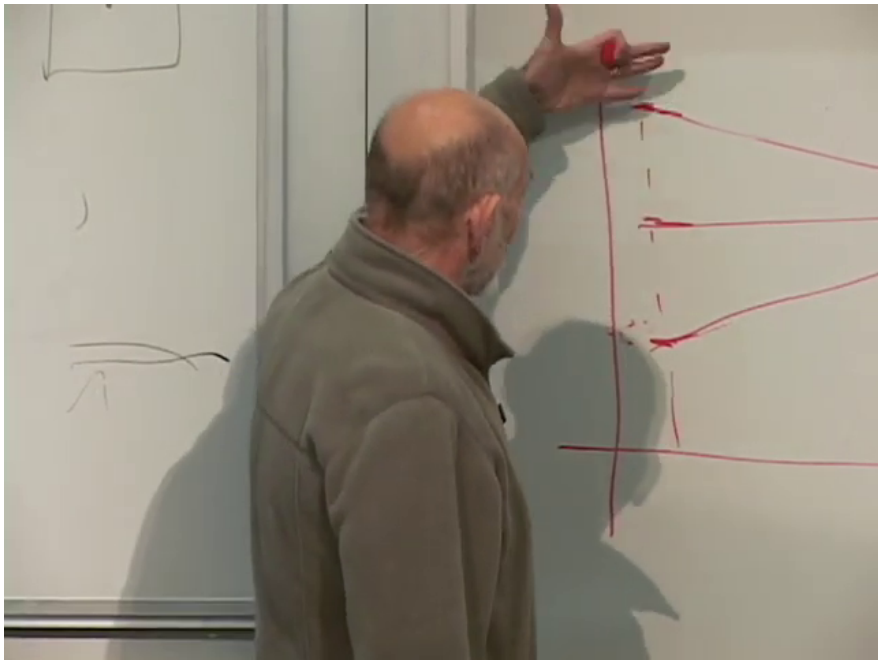


Figure 1.7: The schematic running-coupling plot. The three gauge curves approach one another before the rising gravitational strength becomes order one.

Lecture frame: 00:47:18.

This is an extrapolation, not a direct experiment at the crossing point. Additional particles or thresholds between laboratory and unification energies can change the slopes. The cautious conclusion is only that known field theory shows no sign of string-sized substructure at ordinary distances, and that a plausible string scale lies somewhere in the enormous interval between the inferred unification and Planck scales.

Question from the audience. Which three couplings are being extrapolated? [a]

Response. They are the gauge couplings associated with $U(1)$, $SU(2)$, and $SU(3)$: the hypercharge, weak, and strong interactions. Electromagnetism emerges from electroweak mixing rather than being an independent fourth gauge factor in this accounting.

[a]Lecture timestamp: 00:48:28.

Question from the audience. Does the gravitational constant G itself increase as we probe higher energy? [a]

Response. G is dimensional, so it is not by itself the correct quantity to compare with a dimensionless gauge coupling. At a characteristic energy M, the dimensionless gravitational strength behaves schematically as $GM^2/(\hbar c)$. That combination grows with energy and becomes order one at the Planck scale.

[a]Lecture timestamp: 00:49:30.

The force laws make the comparison visible:

$$F_{\text{em}} \sim \frac{e^2}{r^2}, \qquad F_{\text{grav}} \sim \frac{GM^2}{r^2}. \tag{1.16}$$

Thus e^2 is compared not with G, but with GM^2 in natural units.

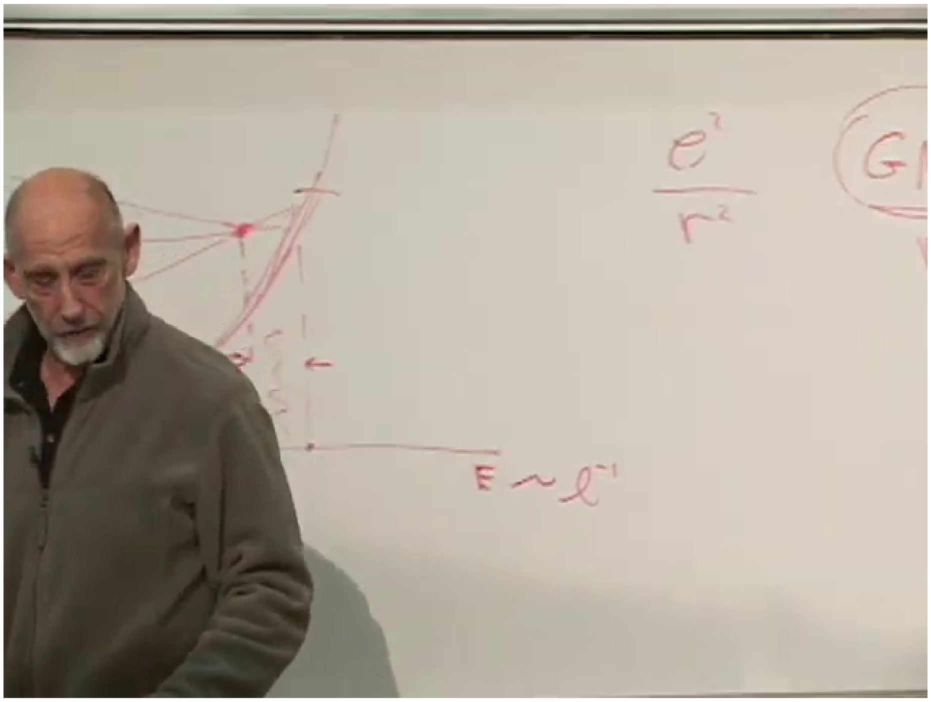


Figure 1.8: Electromagnetic and gravitational force laws used to identify the dimensionless gravitational strength.

Lecture frame: 00:50:50.

1.4 Why Direct String Spectroscopy Is Remote

How would we establish experimentally that an electron is a string? We would do what hadron spectroscopy did: strike it hard enough to excite its internal modes and look for a tower whose masses follow the string pattern. If the first massive level lies near 10^{19} GeV, that is far beyond any direct machine.

The total energy is not the only obstacle. It must be concentrated within a tiny impact parameter. That brings in black holes.

Question from the audience. At these energies would an accelerator begin producing small black holes? [a]

Response. That is the heuristic expectation once center-of-mass energy is concentrated inside its gravitational radius. A stellar black hole is made by gravitational potential energy squeezing a massive star. A microscopic one would instead be made by kinetic energy focused into a tiny collision region. In four-dimensional gravity the required scale is Planckian.

[a] Lecture timestamp: 00:50:57.

For perspective, a linear accelerator's energy gain is roughly proportional to its length. The lecture takes about two miles for roughly 100 GeV as a simple SLAC-scale yardstick. Reaching 10^{19} GeV at the same gradient would multiply the length by roughly 10^{17}, producing a machine of galactic scale.

Even such a machine would need luminosity. Beams are not aimed with Planck-length precision one pair at a time; enormous numbers of particles cross so that a few collide at the desired impact parameter. Multiplying a Planck energy by an enormous collision rate produces an equally absurd power requirement. The estimates are intentionally rough, but their conclusion is stable: direct string spectroscopy at the Planck scale is not an experiment one can plausibly design.

Indirect evidence would have to be more structural: a theory that yields the observed low-energy particles and interactions with enough specificity to be tested, or effects of compactification that survive far below the string scale. The lecture does not claim that such a decisive construction is already in hand.

1.4.1 Extra dimensions and the localization limit

Question from the audience. Should the extra dimensions also have sizes near the Planck length? [a]

Response. They are expected to be extremely small in this elementary picture, though not fixed exactly by the argument. If a compact dimension were large enough to explore at lower energy, new Kaluza–Klein modes and higher-dimensional behavior would already have altered the field theory we observe. Its successful extrapolation therefore places an upper bound on such dimensions.

[a]Lecture timestamp: 00:56:23.

Question from the audience. Why could a string or compact dimension not be many orders of magnitude smaller than ℓ_P? [a]

Response. Resolving a shorter distance requires a larger momentum. Beyond the Planck scale the energy needed for localization has a gravitational radius comparable to, and then larger than, the region being probed. The experiment makes a black hole before it resolves a smaller ordinary string.

[a]Lecture timestamp: 00:57:48.

The uncertainty principle gives the estimate. Localizing to Δx requires

Question from the audience. Once Planck length and Planck time are chosen, how is the Planck mass physically determined? [a]

Response. Use either $\hbar$ or G to convert the length scale into an energy. The uncertainty principle says that localizing two particles within ℓ_P requires momentum of order $\hbar/\ell_P$, hence energy of order $m_P c^2$. Equivalently, it is the mass for which quantum and gravitational lengths become comparable.

[a]Lecture timestamp: 00:59:04.

$$\Delta p \gtrsim \frac{\hbar}{\Delta x}, \qquad E \sim c\,\Delta p. \tag{1.17}$$

At $\Delta x = \ell_P$, this energy is

$$E \sim \frac{\hbar c}{\ell_P} = m_P c^2. \tag{1.18}$$

At the same time the Schwarzschild radius

$$r_s \sim \frac{GE}{c^4} \tag{1.19}$$

is of order ℓ_P, up to numerical constants. Quantum localization and gravitational collapse meet at the same scale.

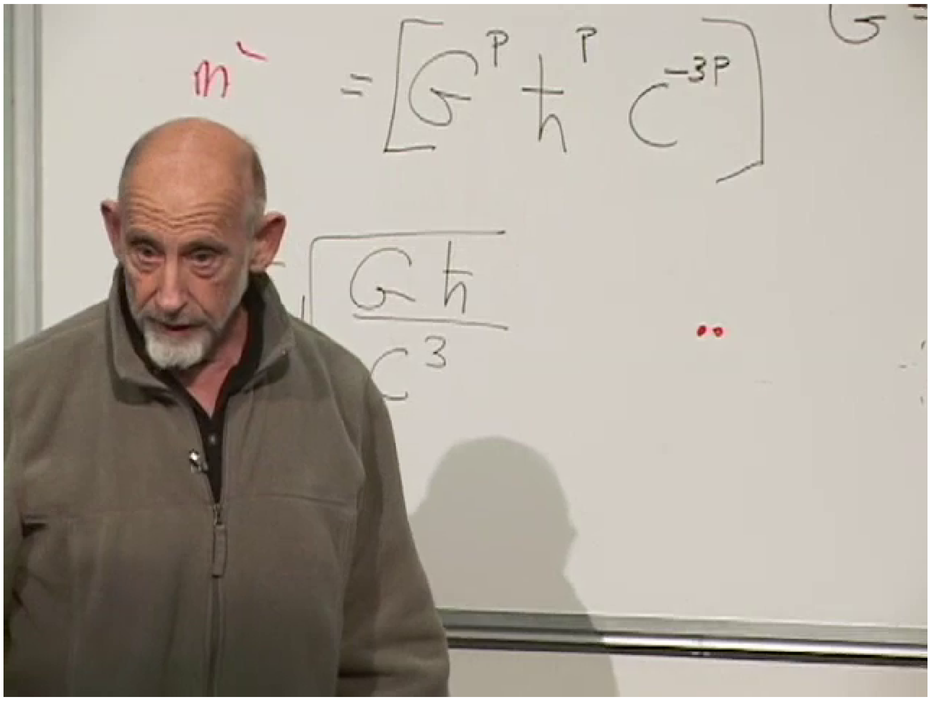


Figure 1.9: The Planck length and mass formulas revisited during the localization argument.
Lecture frame: 01:00:25.

This completes the scale-setting argument.

1.5 The Ground State Must Sit Below the Photon

There are two possible directions after the break: begin conformal transformations, or give the older and easier route to the critical dimension. We choose the latter, with a warning. The easy argument is not easy, and it is not the deepest reason. It is nevertheless correct and close to the historical discovery.

Return to the open bosonic string. Acting once with a transverse creation operator produces two polarization states. Because there is no longitudinal third state, the excitation must be a massless vector. In units where one mode-1 quantum raises m^2 by one,

$$m^2_{\text{ground}} = -1, \qquad m^2_{\text{first}} = -1 + 1 = 0. \tag{1.20}$$

The tachyonic ground state remains a problem of the bosonic theory; it is acknowledged and set aside. The immediate question is where the negative unit in (1.20) comes from.

Every string mode is a harmonic oscillator. With $\hbar = 1$, its vacuum energy is

$$E_{0,n} = \frac{\omega_n}{2}. \tag{1.21}$$

For the string, $\omega_n = n$, so one transverse coordinate contributes formally

$$E_0^{(1)} = \frac{1}{2}\sum_{n=1}^{\infty} n. \tag{1.22}$$

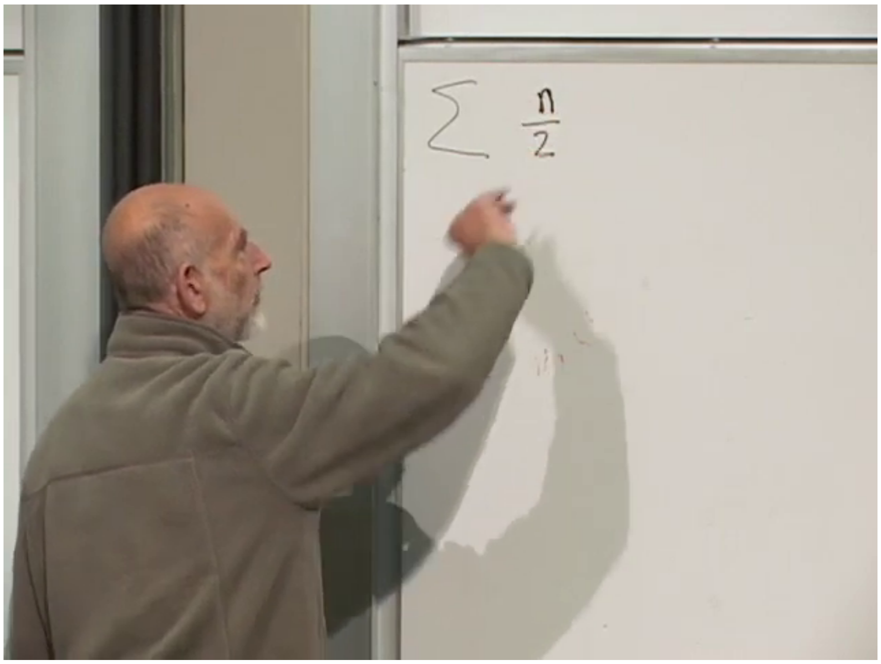


Figure 1.10: The zero-point sum $\sum n/2$ written after the mode frequencies are recalled.
Lecture frame: 01:06:18.

Now the difficulty is sharp. Every term in (1.22) is positive, and the sum diverges. How can it supply a finite negative intercept?

1.6 Subtract the Motion Before Reading the Spectrum

Before regulating the sum, recall why internal energy was identified with mass squared. For a particle carrying a large longitudinal momentum $P > 0$,

$$E = \sqrt{P^2 + m^2} = P + \frac{m^2}{2P} + O(P^{-3}). \tag{1.23}$$

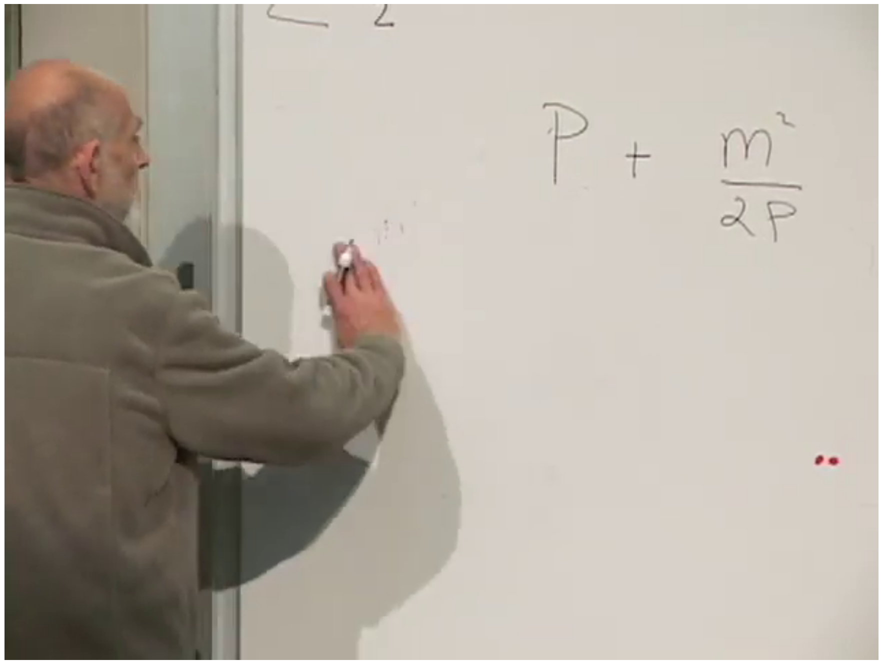


Figure 1.11: The large-P expansion separating overall momentum from the internal mass term.
Lecture frame: 01:07:35.

The first term is the uninteresting overall motion down the z-axis. It is conserved and common to all internal states with the same P. Subtracting it and rescaling removes the kinematic time dilation:

$$\begin{aligned} E - P &= \frac{m^2}{2P} + O(P^{-3}), \\ 2P(E - P) &= m^2 + O(P^{-2}). \end{aligned} \tag{1.24}$$

The atom provides the physical picture. Boost a hydrogen atom. Its total energy grows, but the energy difference between two internal levels, as measured in the laboratory time, shrinks because the atom's clock is dilated. The factor $1/P$ in (1.23) is that suppression. Multiplying by $2P$ restores the invariant mass-squared difference.

The raw ground-state energy therefore need not be literally -1. It may contain a large state-independent piece associated with the overall longitudinal energy, plus a finite internal remainder. The lecture rewrites the boosted relation to display both pieces before turning back to the oscillator sum.

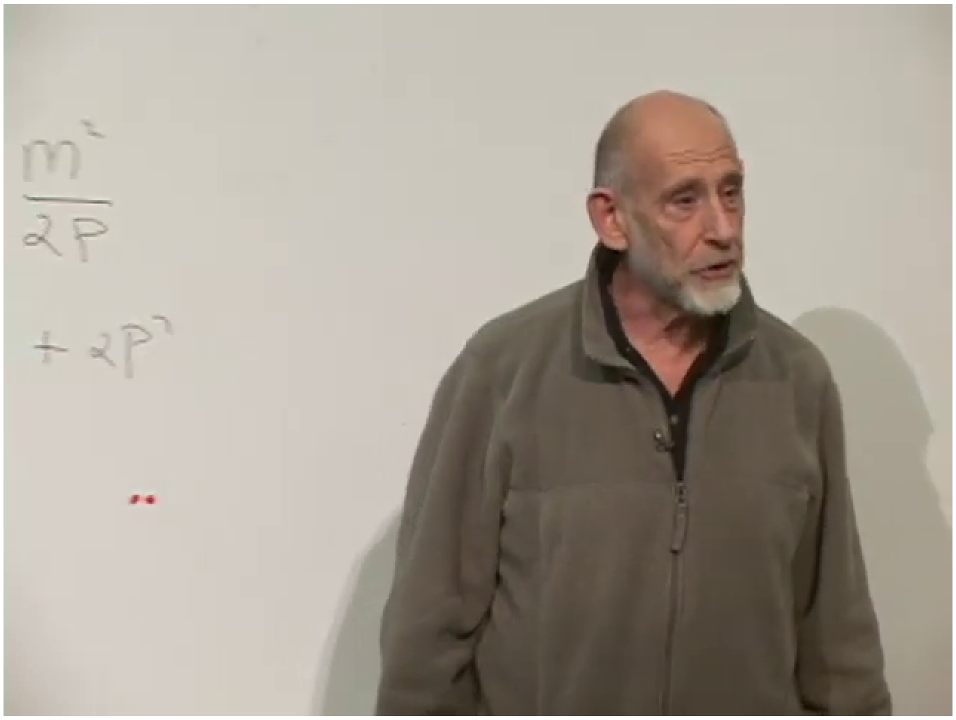


Figure 1.12: The invariant mass term displayed beside the large state-independent longitudinal contribution.

Lecture frame: 01:11:34.

Infinite vacuum energies are familiar in quantum field theory. One often separates a divergent state-independent constant from observable energy differences. The lecture compares this with the zero-point energy of the electromagnetic field in the room and openly calls the procedure unsatisfactory but standard. Here we follow the same heuristic: isolate the divergent part and ask whether a definite finite remainder survives.

1.7 Regulating the Sum

Introduce a positive regulator ϵ:

$$S_1(\epsilon) = \sum_{n=1}^{\infty} n e^{-n\epsilon}. \tag{1.25}$$

For every $\epsilon > 0$, the exponential eventually defeats the factor n, so the sum converges. Sending ϵ toward zero removes the cutoff.

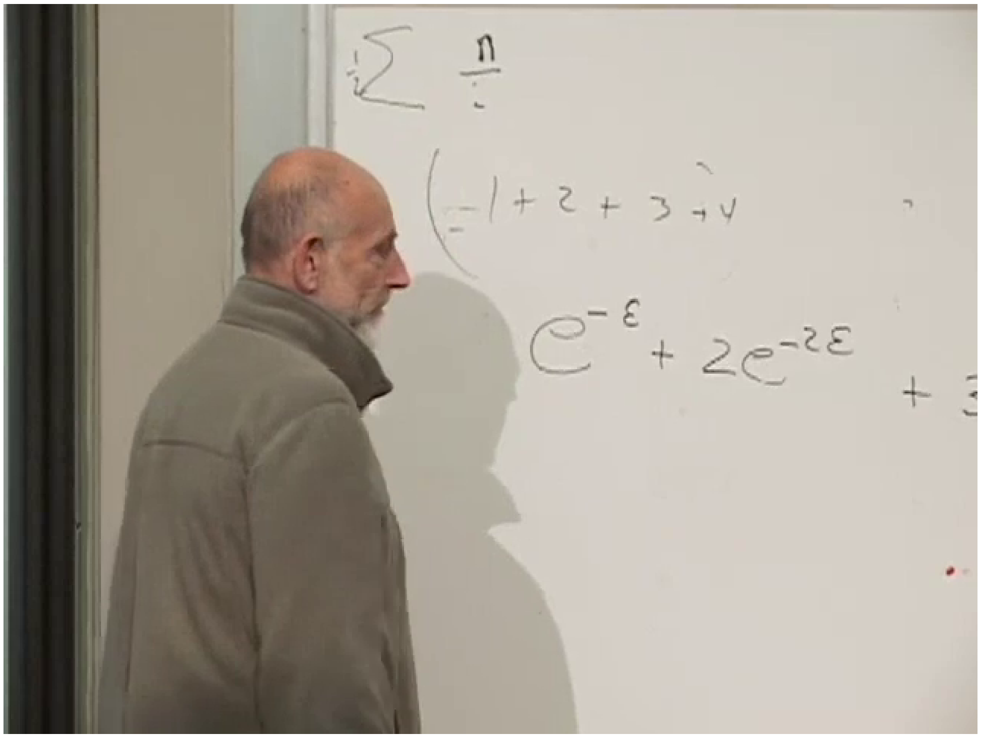


Figure 1.13: The exponential regulator applied term by term to $1 + 2 + 3 + \cdots$.
Lecture frame: 01:16:42.

The factor n can be generated by differentiation:

$$S_1(\epsilon) = -\frac{d}{d\epsilon} \sum_{n=1}^{\infty} e^{-n\epsilon}. \tag{1.26}$$

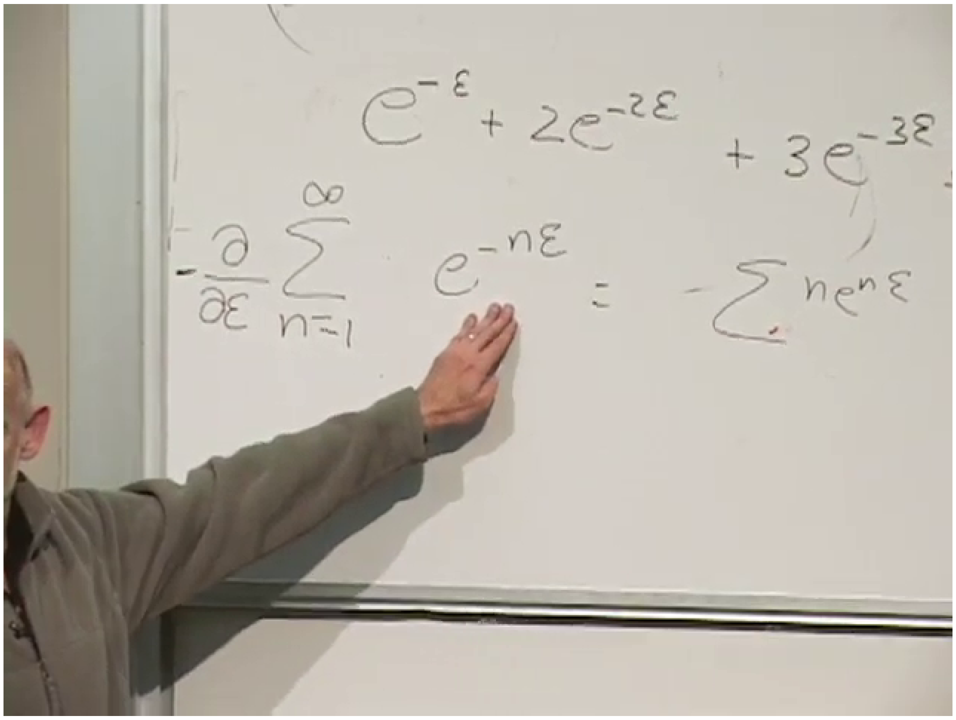


Figure 1.14: Differentiation with respect to the cutoff brings down $-n$.
Lecture frame: 01:18:05.

The remaining series is geometric:

$$S_0(\epsilon) \equiv \sum_{n=1}^{\infty} e^{-n\epsilon} = \frac{e^{-\epsilon}}{1-e^{-\epsilon}} = \frac{1}{e^{\epsilon}-1}. \tag{1.27}$$

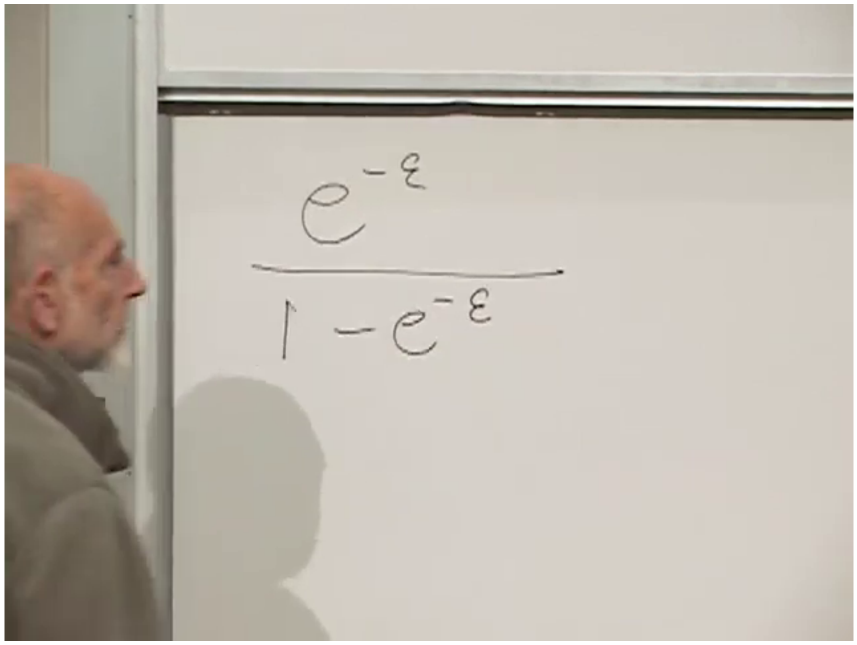


Figure 1.15: The regulated geometric series in closed form.
Lecture frame: 01:19:12.

Expand at small ϵ:

$$S_0(\epsilon) = \frac{1}{\epsilon} - \frac{1}{2} + \frac{\epsilon}{12} - \frac{\epsilon^3}{720} + O(\epsilon^5). \tag{1.28}$$

The board obtains this by expanding numerator and denominator separately, factoring one power of ϵ, and then expanding the reciprocal. That calculation contains several in-class sign checks; the compact expression (1.28) is its cleaned result.

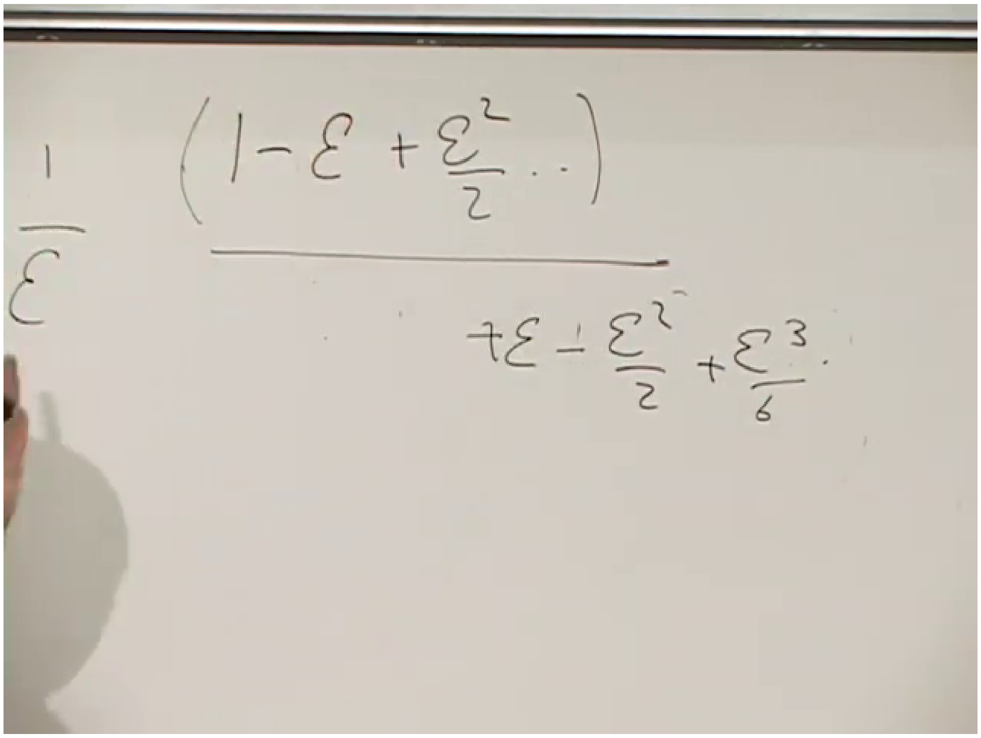


Figure 1.16: The numerator and denominator expansions before the reciprocal is simplified.
Lecture frame: 01:23:10.

Applying $-d/d\epsilon$ gives

$$S_1(\epsilon) = \frac{1}{\epsilon^2} - \frac{1}{12} + \frac{\epsilon^2}{240} + O(\epsilon^4). \tag{1.29}$$

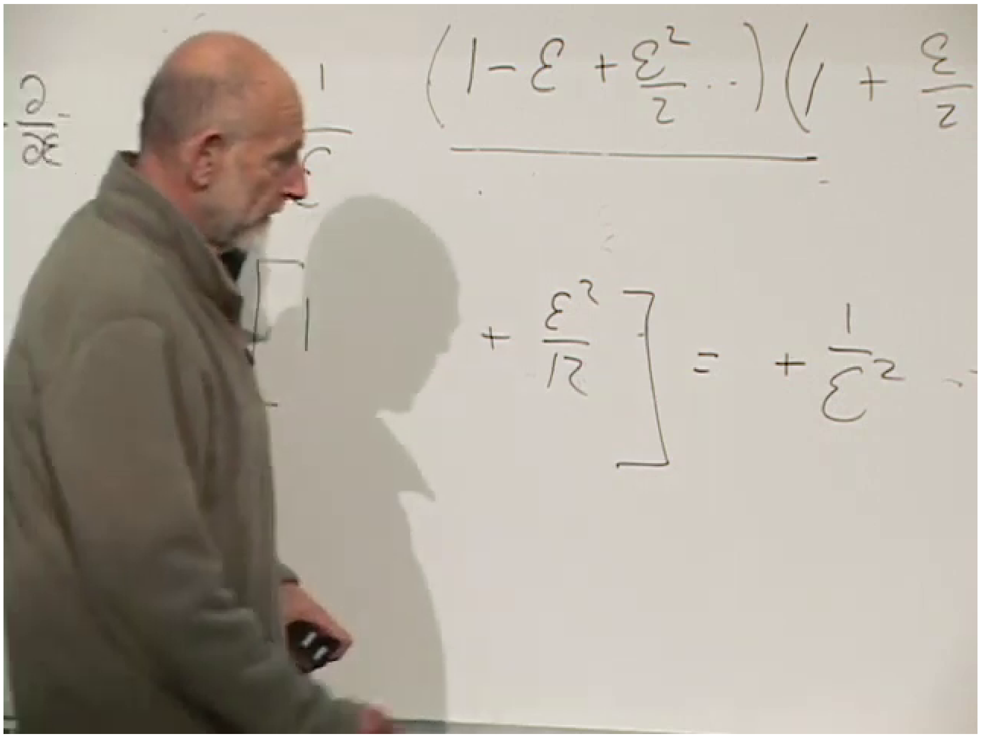


Figure 1.17: The divergent $1/\epsilon^2$ term and finite $-1/12$ remainder at the end of the regulator calculation.

Lecture frame: 01:29:48.

It is common to summarize this result jokingly as $1 + 2 + 3 + \cdots = -1/12$. That is not ordinary convergence. Equation (1.29) says that this particular regulator produces a divergent term plus a regulator-independent finite part:

$$\underbrace{\frac{1}{\epsilon^2}}_{\text{divergent}} + \underbrace{\left(-\frac{1}{12}\right)}_{\text{finite}} + O(\epsilon^2). \tag{1.30}$$

The lecture absorbs the first term into the already-present state-independent piece and retains the second. It also preserves the warning attributed to Pauli: being infinite does not by itself make a quantity zero. A consistent theory must explain why the subtraction is allowed.

Restoring the factor of one-half in (1.22), one transverse coordinate contributes

$$E_0^{(1)} = -\frac{1}{24}. \tag{1.31}$$

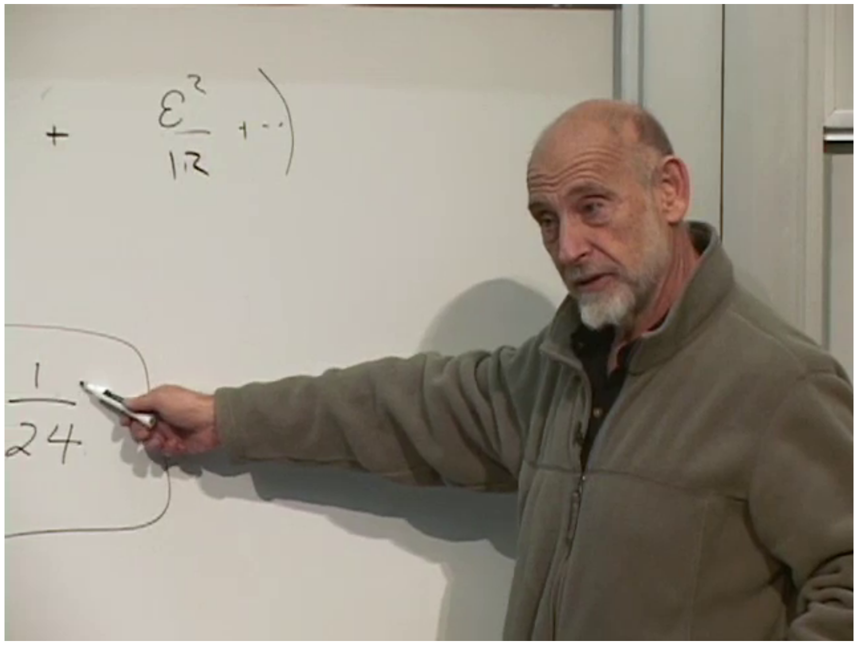


Figure 1.18: The factor of one-half converts the finite $-1/12$ into $-1/24$ per transverse coordinate.
Lecture frame: 01:33:42.

1.8 Counting the Transverse Directions

Let $D_\perp$ be the number of transverse string coordinates. Their total vacuum contribution is

$$E_0^{\text{open}} = -\frac{D_\perp}{24}. \tag{1.32}$$

For the first open-string oscillator to raise the ground state from -1 to the massless level, we need

$$-\frac{D_\perp}{24} = -1. \tag{1.33}$$

Therefore

$$D_\perp = 24. \tag{1.34}$$

Those are the directions perpendicular to the large longitudinal momentum. Adding that longitudinal spatial direction and time gives

$$D = D_\perp + 2 = 26. \tag{1.35}$$

Question from the audience. When did this argument enter string theory? [a]

Response. It belongs to the original hadronic-string era, around 1969 and the early 1970s. Regge trajectories were already known experimentally. Once the oscillator model was explored on its own, the first excited state was seen to have two rather than three vector polarizations, forcing it to be massless. The required intercept then led quickly to twenty-four transverse oscillators.

[a] Lecture timestamp: 01:35:27.

At the time this looked almost too strange to believe. Why should a regularized vacuum sum know the dimension of spacetime? Yet the pieces fit: the massless vector, the intercept, and the transverse oscillator count all selected the same number.

1.8.1 The closed string checks the count

A closed string needs one left-moving and one right-moving mode to reach its first physical massless level. Its ground state must therefore begin two units below zero.

Question from the audience. If twenty-four transverse directions give only -1, must a closed string live in forty-eight transverse directions? [a]

Response. No. The same spacetime contains both open and closed strings. A closed string already has twice the oscillator content: one tower circulating in each world-sheet direction. Each tower contributes $-24/24 = -1$, so together they give -2.

[a] Lecture timestamp: 01:37:47.

Thus

$$E_0^{\text{closed}} = 2\left(-\frac{D_\perp}{24}\right) = -2 \qquad (D_\perp = 24). \tag{1.36}$$

Adding one oscillator from each sector raises the state by two units and lands at zero mass squared, exactly where the graviton multiplet appeared in the preceding lecture.

This agreement makes the argument smell better, but it is still not the deep explanation. In modern quantization the normal-ordering constant, closure of the Lorentz or Virasoro algebra, and cancellation of the conformal anomaly belong to one consistent structure. That structure fixes both the intercept and the critical dimension; the finite cutoff remainder is the historical way into it. [3]

That deeper route is next. Only a little complex analysis is needed to begin: complex coordinates, analytic functions, and the Cauchy–Riemann equations. The promised physical image is Turkish taffy—pulled one way, stretched another, reshaped without destroying the structure that matters. The mathematics behind that image is conformal invariance.

[3] Editorial clarification: The regulator calculation is a heuristic view of the normal-ordering constant. The deeper derivation of $D = 26$ requires world-sheet conformal invariance, equivalently cancellation of the conformal anomaly and consistent constraint algebra.